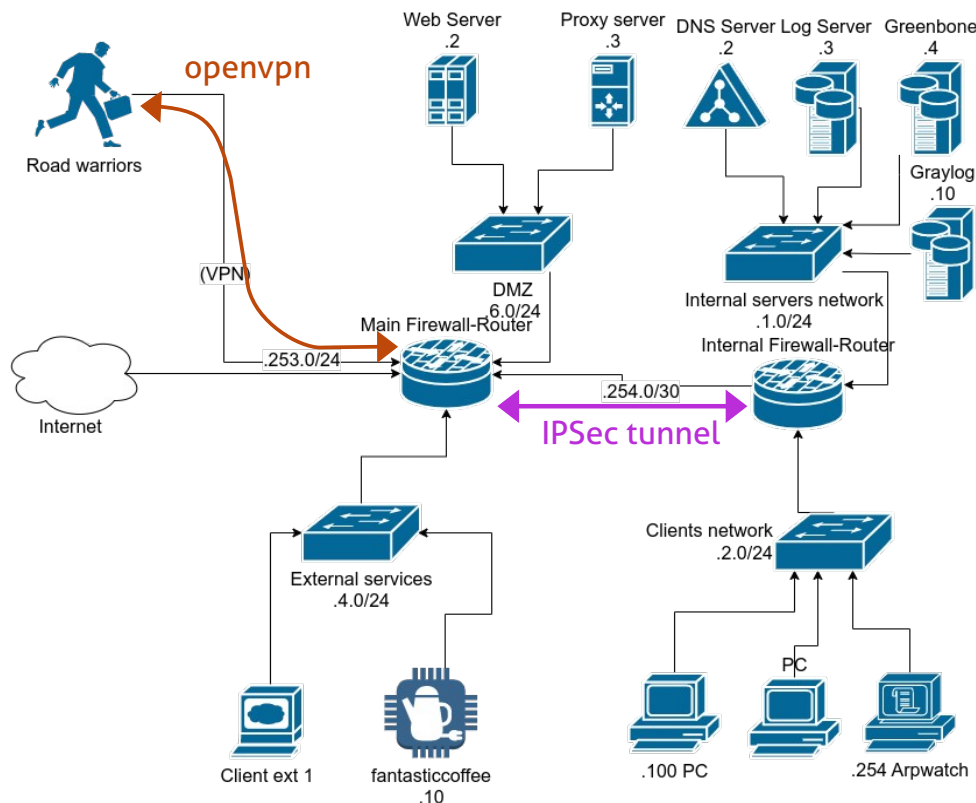


Assignment 2 (v.1.0)¹: VPN on ACME co.

In this assignment, you have to properly configure two new services running on the ACME co.'s networks. In particular, a VPN for the road warriors employee and a VPN tunnel between the Main and the Internal routers (running on two different branch offices). The two VPNs have to work with IPv4, whereas IPv6 is optional.

The VPN for the road warriors has to be implemented with the `openvpn` option (or similar), while the internal tunnel must be implemented with IPSec.

ACME co. TOPOLOGY



Road-warriors VPN requirements

You must choose the most appropriate configuration to allow the road warriors of ACME Co. to join the company networks and access the internal services. You must choose where to place the VPN gateway and adequately configure the network to enforce the VPN requirements. The road warriors should have access from the Internet, using the public IP address assigned to the WAN interface of the Main-firewall. The VPN must allow road warriors to access the company's internal networks.

The ACME Co.'s network has four types of road-warrior users: **External User**, **Standard User**, **Power User**, and **Administrator**. External users can access only the external services network; a standard user can access only the DMZ network; power users can access both the DMZ and the server networks; and an administrator can access all the company's networks. The authentication mechanism must rely on a Public Key Infrastructure (X.509).

You have to create the following users:

- Alice, external user;
- Bob, standard user;
- Christina, power user;
- Diana, administrator.

Main-Internal VPN tunnel

You have to properly setup a VPN tunnel between the two routers, since the actual connection is running on an insecure medium: any packet running between the two routers should go through an **IPSec tunnel**. You can decide the security level and the implementation details, motivating your choices.

Scheme of your hand-in

You have to prepare a document that reports the activity you have performed to realize the assignment. When uploading your report in classroom, you have also to include the details about how you tested the VPN connections of the four accounts above mentioned and the site-to-site VPN.

The document should be named `ACME_XX_a2_report.pdf` and should be included together with the exported firewall configurations (and any other relevant configuration file to test the VPNs) in a .zip file named `ACME_XX_a2.zip`. The zip file is **the only thing** your group has to hand-in in classroom (only one of the members). The document should be clear and concise (few pages, please...) and have at least the following pieces of information:

1. Group number
2. Student names and numbers
3. Initial brainstorming, where you write your considerations and observations about what to do and how. Here you have to motivate the choice you made for the whole setup.
4. Details about the VPN setup for the road warriors (namely, the installation procedure, together configuration files, if any, and account setup).
5. Details about the VPN for the Main-Internal VPN tunnel.
6. Test of the new configuration.
7. Final remarks.

Hand-in dates

The assignment has to be handed-in before taking the written exam. The evaluation will be done considering both the completeness, clearness and proficiency.